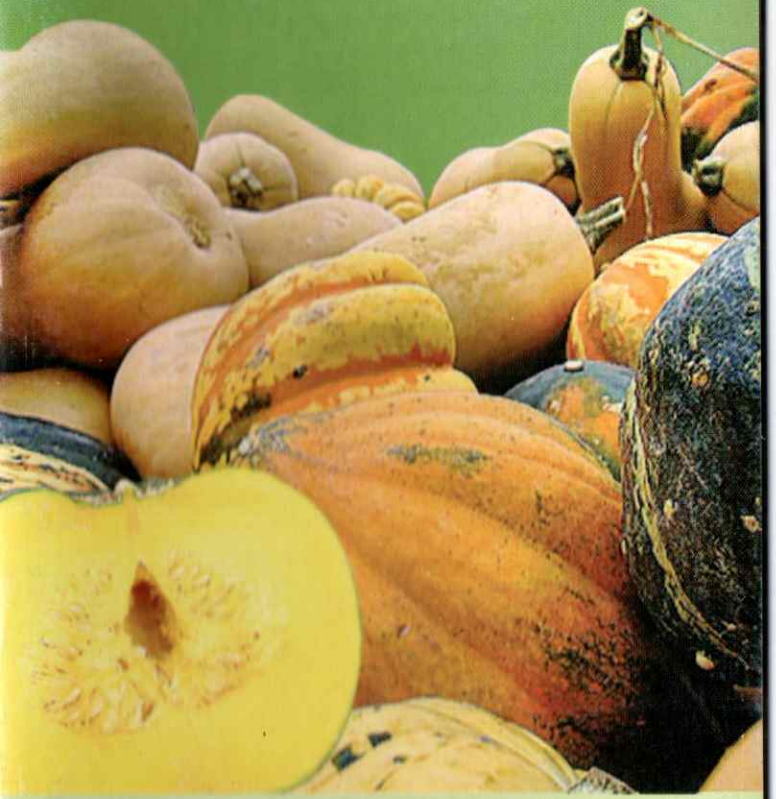
ISSN 1908-5710

INFORMATION, EDUCATION AND
COMMUNICATION MATERIALS SERIES NO. 7-6

Republic of the Philippines
Department of Agriculture
AGRICULTURAL TRAINING INSTITUTE

Republic of the Philippines
DEPARTMENT OF AGRICULTURE
REGIONAL FIELD UNIT NO. 02



SQUASH
Production Guide

Squash *(Cucurbita maxima L.)* locally known as "kalabasa" is cultivated throughout the Philippines. It is a popular vegetable among the Filipinos. The shoots, flowers and young fruits can be cooked as *dinengdeng*, *bulanglang* or sauteed with meat, fish or shrimps and mixed with other vegetables. Aside from these uses, squash fruits can also be made into catsup, pastillas, hopia, leche flan, flour for bread, chips and curls. The seeds can also be made into kutchi. Like other cucurbits, squash is recognized as an excellent source of vitamins (A, C, B_1, B_6, B_3) and minerals. Beta carotene contained in squash is a powerful anti-oxidant and anti-inflamatory. It also prevents the oxidation of cholesterol is on the body. Potassium is also present in squash. It reduces urinary calcium excretion.

SOIL AND CLIMATIC REQUIREMENTS

Squash thrives in any type of soil however, it performs best in well-drained sandy loam soil with a pH level of 5.5 to 6.5. The crop requires a relatively dry warm climate which is essential for fruit setting. The optimum temperature for good crop growth ranges from 18 to 30°C. For optimum yield, plant the crop during October to December in lowland areas while May to July in highland areas.

CULTURAL MANAGEMENT PRACTICES

Selection of Varieties

For better yield and profit, select varieties that are adaptable to local conditions, market preference and resistant to insect pests and diseases. To guide you in selecting the right varieties, refer to the "*Guide in Selecting Lowland Vegetable Varieties*" included in this kit.

Land Preparation

Plow and harrow the field 2 to 3 times at one week interval to prevent growth of weeds. After the last harrowing, make furrows with distance depending on the season. For rainy season, the distance of furrows is 3 meters. During dry season, the furrows should be 0.5 meters

Construct drainage canals to prevent water logging during rainy season.

Zero tillage can also be applied in squash production. Before planting, clear the area by cutting the weeds manually or by spraying of herbicides. Dig holes at a depth of 10 cm with a distance of 3m x 3m. Pulverize the soil where the seeds will be sown.

Planting

Direct Planting

- Plant one pre-germinated seed per hill at a distance of 1 meter between hills.
- Cover the seeds with thin layer of soil.
- During wet season, plant in ridges or above furrows to prevent rotting of seedlings due to flooding.

Transplanting

- Use a prepared media of one part compost or organic fertilizer, one part clay soil and one part carbonized rice hull. A ready mixed commercial soil media for seedling production can also be used.
- Fill in plastic bags, potlets or seedling trays with the prepared media.
- Water the potting media before sowing. Sow one pre-germinated seed per potlet.
- Place the seedling trays/ potlets under a temporary shade.

- Maintain the seedlings by watering regularly when needed.
- Harden the seedlings by gradually reducing the frequency of watering and exposing to direct sunlight.
- Transplant one seedling per hill at a distance of 1 m between hills 15 days after emergence or when true leaves have developed.
- Transplant in the afternoon or during cloudy days.
- Replant missing hills immediately.

Fertilizer Application

The kind and amount of fertilizer to be applied depends on the result of soil analysis. In the absence of soil analysis, apply the following:

- For lowland areas, incorporate fully decomposed farm manures or compose fertilizer at harrowing, before planting. For highland areas, apply 100 g organic fertilizer mixed with 1 tbsp of complete fertilizer in each hole before planting.
- Sidedress 1 tbsp of urea per hill one month after planting or when the plants have produced vines about 1 meter long.
- Two weeks after the first sidedressing, apply 10 g of potash (0-0-60) and 60 g of solophos (0-60-0) per plant.
- Additional urea or potash may be applied every 15 days depending on crop stand.

Irrigation

Water is very important for the whole duration of the crop especially during the fruiting stage.

Furrow irrigation is preferable in large scale production. For backyard gardening, use sprinkler or any modified system of irrigation.

Irrigate once a week or hand water with one liter per hill every other day. Irrigate also every after fertilizer application. Withhold watering or irrigation when the fruits have reached mature green stage. Irrigating at this stage will result in lower fruit quality.

Weeding

Remove weeds in between plants at 7 to 14 days after emergence. Repeat weeding if weeds still persist.

Crop Protection

Aphids and 28-spotted lady beetle are the most prevalent insect pests of squash throughout the year. For diseases, powdery mildew and mosaic virus are the most common but the most damaging is mosaic virus. The suggested management strategies for pests and diseases are presented in the *Insect Pest and Disease Management Guide*

Hand Pollination

To increase fruit setting when insect pollinators are few, hand pollinate by inserting the male flower of the same age to the female flower between 6:00 and 8:00 in the morning.

Thinning of Vines and Fruits

Thin some lateral vines near the fully developed fruits. Remove all deformed fruits while still small to avoid nutrient competition. Place 5 to 6 cm thick of bedding materials like rice straw, rice hull, etc. beneath the good fruits to prevent rotting of the fruits.

Harvesting

Harvest immature fruits at 30 to 40 days from pollination. Cut the peduncle using a knife with a portion still attached if it cannot be marketed immediately to avoid fruit damage.

If the fruits are intended for seed purposes, the appearance of powdery, whitish substance on the surface of the fruit, and the hardening of the rind are indices of a fully matured fruits. At such stage, the seeds of squash are already matured and fully developed. Harvest the matured fruit when the fruits appear brownish orange. Harvest fruits of plants that have no symptoms of viral infection, (mosaic, yellowing of leaves) if intended for seeds. This must be done to avoid virus-infected seeds for the next crop.

COST AND RETURN ANALYSIS FOR SQUASH

Per Hectare Basis for CY 2008

PARTICULARS	UNIT	QNTY.	UNIT COST (P)	TOTAL COST (P)
A.Labor Inputs/Hectare				
1. Land Preparation				
a. 1st Plowing		8	300.00	2,400.00
b. 1st Harrowing		6	300.00	1,800.00
c. 2nd Plowing		6	300.00	1,800.00
d. 2nd Harrowing		4	300.00	1,200.00
f. Furrowing		3	300.00	900.00
2. Planting	MD	8	150.00	1,200.00
3. Fertilization application				
-1st application	MD	4	150.00	600.00
-2nd application	MD	4	150.00	600.00
-3rd application	MD	4	150.00	600.00
4. Irrigation/Watering (5x)	MD	5 x 4	150.00	3,000.00
5. Weeding and Cultivation				
a. Hand weeding (2x)	MD	2 x 6	150.00	1,800.00
b. Hilling up	MD	4	150.00	600.00
6. Control of Pest & diseases (6x spraying)	MD	18	150.00	2,700.00
7. Harvesting (3x)	MD	3 x 4	150.00	1,800.00
8. Sorting and packing (3x)	MD	3 x 2	150.00	900.00
9. Hauling (3x)	MD	3 x 2	150.00	900.00
Sub-Total				**22,800.00**
B.Materials Inputs				
1. Seeds (OPV)	kg	2	1,000.00	2,000.00
2. Fertilizer				
- Complete (14-14-14)	bag	2	1,900.00	3,800.00
- Urea (45-0-0)	bag	1	2,000.00	2,000.00
- Muriate of Potash	bag	1	1,000.00	1,000.00
- Organic Fertilizer	bag	20	200.00	4,000.00
3. Insecticide	kg	4	850.00	3,400.00
4. Fungicide	kg	2	780.00	1,560.00
5. Fuel	liter	20	50.00	1,000.00
6. Jute Sack	sack	600	5.00	3,000.00
7. Miscellaneous (straw, nail 3, needles, etc.)				1,500.00
SUB-TOTAL				**23,260.00**
Sub-TOTAL (A&B)				**46,060.00**
C Contingencies (15% of the total labor & material inputs)				**6,909.00**
GRAND TOTAL				**52,969.00**
Estimated Yield/ha (kg)				**30,000.00**
Estimated Gross Income				**300,000.00**
Estimated Net Income				**247,031.00**
Return on Investment (ROI)				**466%**
Break even Price (Per Kilo)				**1.80**

MD-man days

REFERENCES

TECHNOPACKAGE FOR UPO. Agricultural Pilot Center, DA-CVLMROS, Iguig, Cagayan

GINTONG ANI PRODUCTION GUIDE FOR SQUASH. Department of Agriculture Regional Field Office No. 02, Tuguegarao City.

COST OF PRODUCTION PER HECTARE OF DIFFERENT VEGETABLES: ISABELA CONDITION. 2004. A Handout. HVCC Section, Department of Agriculture Regional Field Unit 02

SQUASH PRODUCTION. 2008. PHILRICE Publication. www.open seedomy.ph